12. Which of the following statements is not true?

a) on passing H_2S, through acidified $K_2Cr_2O_7$ solution, a milky colour is observed.

b) $Na_2Cr_2O_7$ is preferred over $K_2Cr_2O_7$ in volumetric analysis

c) $K_2Cr_2O_7$ solution in acidic medium is orange in colour

d) $K_2Cr_2O_7$ solution becomes yellow on increasing the P^H beyond 7

13. Permanganate ion changes to _____ in acidic medium

a) MnO_4^{2-} b) Mn^{2+}

c) Mn^{3+} d) MnO_2

14. A white crystalline salt (A) react with dilute HCl to liberate a suffocating gas (B) and also forms a yellow precipitate . The gas (B) turns potassium dichromate acidified with dil H_2SO_4 to a green coloured solution(C). A,B and C are respectively

a) Na_2SO_3 , SO_2 , $Cr_2(SO_4)_3$ b) $Na_2S_2O_3$, SO_2 , $Cr_2(SO_4)_3$

c) Na_2S , SO_2 , $Cr_2(SO_4)_3$ d) Na_2SO_4 , SO_2 , $Cr_2(SO_4)_3$

15. MnO_4^- react with Br^- in alkaline P^H to give

a) BrO_3^- , MnO_2 b) Br_2, MnO_4^{2-}

c) Br_2, MnO_2 d) BrO^-, MnO_4^{2-}

16. How many moles of I_2 are liberated when 1 mole of potassium dichromate react with potassium iodide?

a) 1 b) 2

c) 3 d) 4

17. The number of moles of acidified $KMnO_4$ required to oxidize 1 mole of ferrous oxalate(FeC_2O_4) is

a) 5 b) 3 c) 0.6 d) 1.5

18. When a brown compound of Mn (A) ids treated with HCl , it gives a gas (B) . The gas (B) taken in excess reacts with NH3 to give an explosive compound (C). The compound A, B and C are

a) MnO_2, Cl_2 , NCl_3 b) MnO, Cl_2 , NH_4Cl

c) Mn_3O_4, Cl_2 , NCl_3 d) MnO_3, Cl_2 , NCl_2

19. Which one of the following statements related to lanthanons is incorrect?

a) Europium shows +2 oxidation state.

b) The basicity decreases as the ionic radius decreases from Pr to Lu.

c) All the lanthanons are much more reactive than aluminium.

d) Ce^{4+} solutions are widely used as oxidising agents in volumetric analysis.

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